Resilient Communication Networks

Resilient Behavior and Metrics



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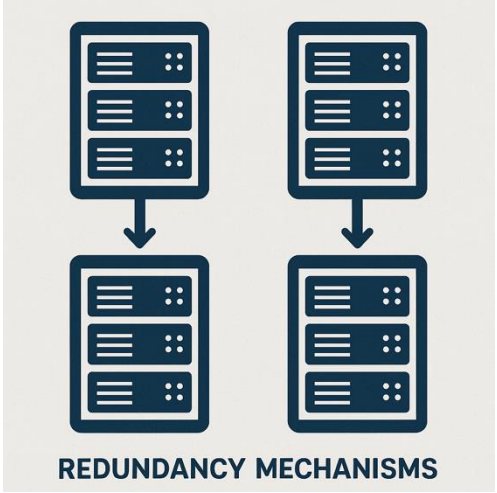
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Enablers of Resilience in Communication Networks

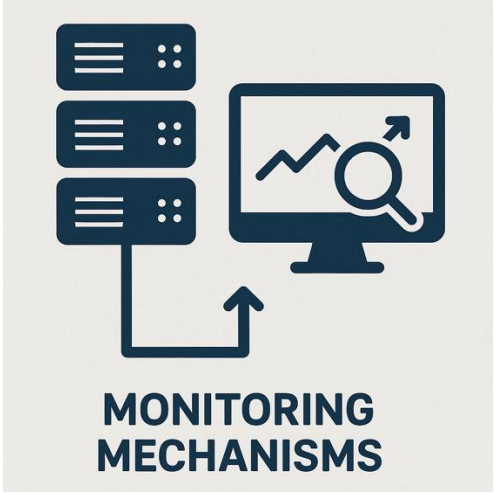
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Four general classes of enablers

- Redundancy for robustness against disruptions
- Protection for robustness against attacks
- Monitoring for detecting disruptions
- Recovery from failures

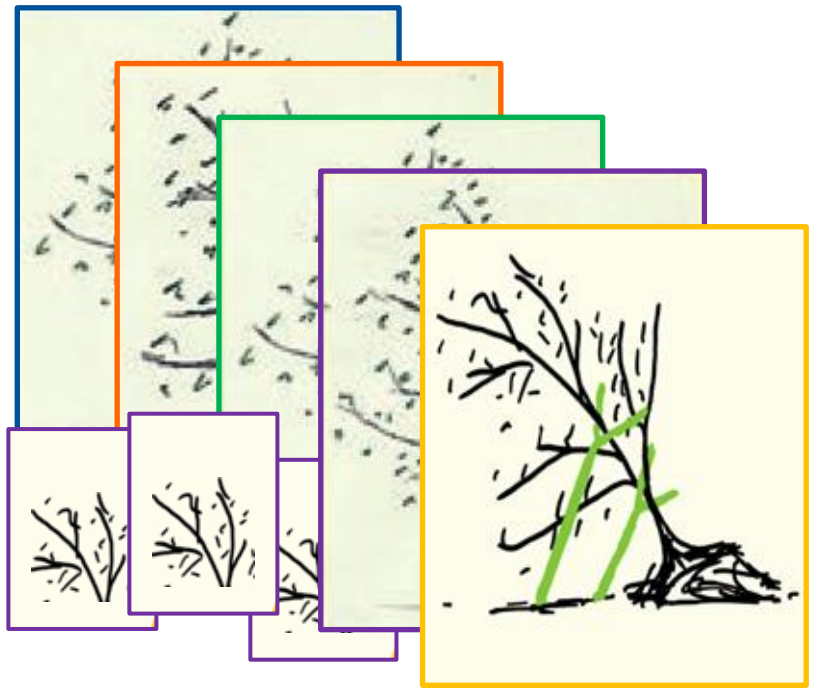








Resilience in Communication Networks

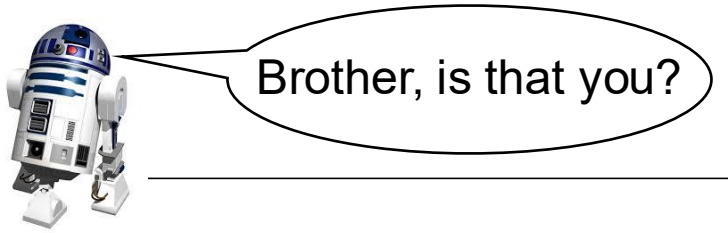


- Absorption through resistance, elasticity and adaptability
- Recovery by returning to normal state
- Transformation through the transfer of functions
- Sustainability through learning from experience and maturation / further development

Resilience describes the ability of a system

(1) to absorb crises,
(2) to recover from them quickly and sustainably,
(3) to achieve comparable or new functionality through transformation

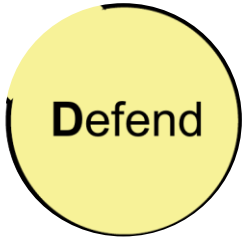
The ResiliNets Strategy: D^2R^2 + DR



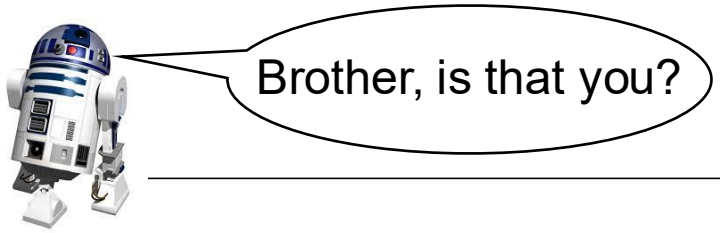
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Defend against challenges and attacks

- For example, using firewalls, redundancy mechanisms, …
- First line of defense against challenges, i.e., network robustness



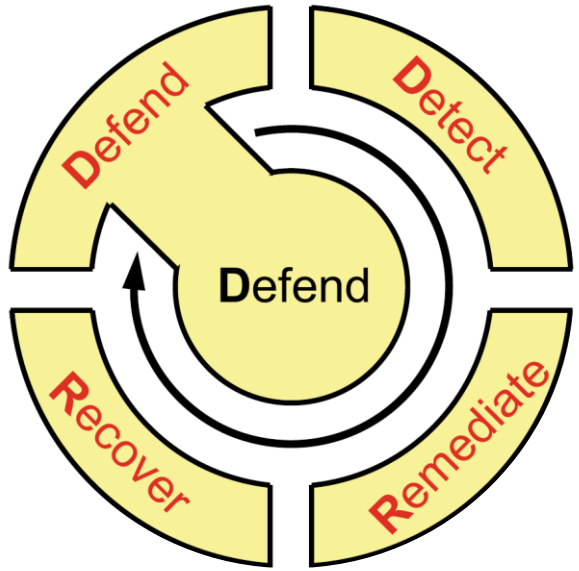
The ResiliNets Strategy: D^2R^2 + DR



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Inner loop

- Passive defense and real-time steps
- Set of activities to rapidly adapt to challenges and attacks and maint
- Triggered when adverse event or condition is detected



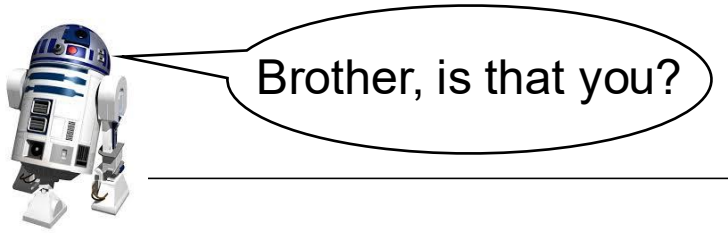
Defend
- against attacks/ challenges

Detect
- disturbances by observing the network

Remediate
- possible disturbances

Recover
- in case of failure

The ResiliNets Strategy: D^2R^2 + DR



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Outer loop

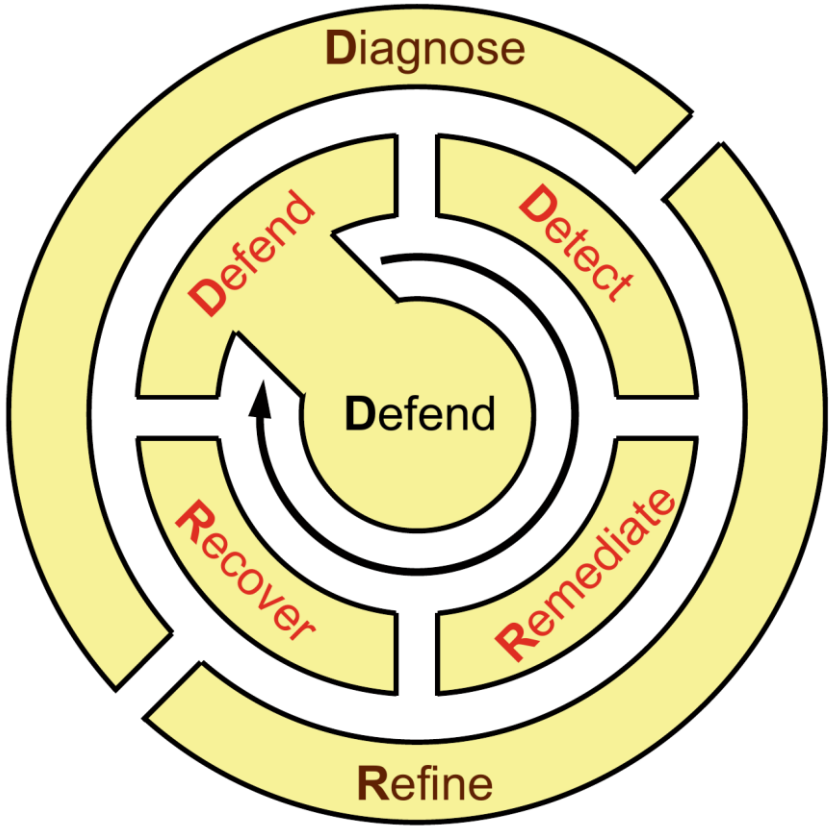
- Enables long-term evolution to enhance the inner loop
- Observe and modify the behavior of the inner loop
- Can be AI-driven or based on human analysis

Diagnose

- the fault

Refine

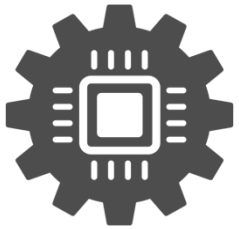
- the system



Resilient Behavior



Self-Organizing & Autonomic Behavior
Initializing



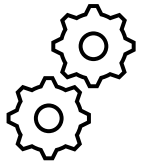
- **Auto-Configuration**
- **Self-Organization**

Setup and **configuration** of the network

Deployment of network hardware and software

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Auto-Configuration



Automatically **configure** the network with suitable parameters and policies

Examples:

- DHCP (Dynamic Host Configuration Protocol)
- BGP (Border Gateway Protocol)

Self-Organizing & Autonomic Behavior
Initializing

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Self-Organization



Dynamically **adapt and reorganize**
in response to changes in network conditions
w/o human intervention

Examples:

Dynamic Routing

BGP automatically discovers and adapts to changes

Load Balancing

Distributes network traffic across multiple paths ensuring equal distribution

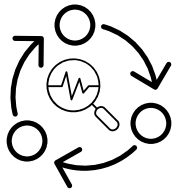
Fault Tolerance

Enables networks to continue operating even during failures

Self-Organizing & Autonomic Behavior
Steady-State Normal

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Self-Managing



Autonomously **monitor** and **manage** own operation

Examples:

- Performance monitoring (e.g., packet loss rates)
- Resource allocation (e.g., bandwidth and power)
- Security management (e.g., intrusion detection and prevention)

Self-Organizing & Autonomic Behavior
Steady-State Normal

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Self-Optimization



Automatically **adjust** own behavior
to **optimize network performance**

Examples:

- Traffic engineering (e.g., minimize latency, maximize bandwidth utilization)
- Quality of Service (e.g., packet priority)
- Energy efficiency (e.g., energy-saving hardware)

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Self-Diagnose



Detect and **diagnose** faults or problems within the network

Examples:

- Fault detection (e.g., failed components, software bugs)
- Fault isolation (e.g., network segmentation, virtualization)
- Root cause analysis (e.g., using system logs)

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Self-Repair

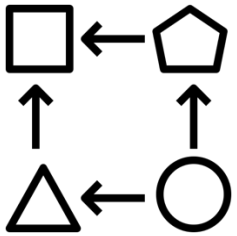


Repair faults or problems within the network

Examples:

- Automatic failover (e.g., redirect traffic)
- Dynamic reconfiguration (e.g., forwarding tables)
- Patching or software updates (e.g., auto-update to patched version)

Adaptability



Design network to operate in dynamic and **constantly changing** network conditions

Defend, **detect**, and **respond** to environment

- Changes in traffic patterns
- Presence of new devices

Remediate challenges

- Redirect traffic
- Adjust network configuration

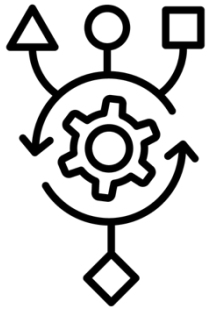
Recover to normal operations

- Reroute traffic
- Activate redundant resources

Evolvability

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Adapt and **evolve over time** in response to emerging threats, new application demands and changing conditions



Refine future behavior to improve response to challenges

- Analyze past events and the effectiveness of network's response
- Identify areas of improvement and appropriate adjustments of network components to detect and remediate future events

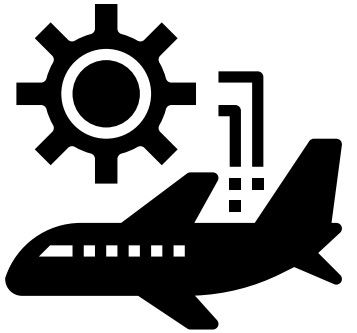
Evolution and **extension** of network architecture

- Adapt to unique requirements of emerging applications
- Technology trends
- Changes in communication environment
- New attack strategies

Evolvability
Refinement of Future Behavior in Aviation

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Goal: Continuously learn from past events and use that knowledge to improve aviation safety



Aviation safety issues prompt investigators to conduct an analysis to **determine what went wrong and how it can be prevented** in the future

- Reflect on the plane's defense against issues
- Detect the issue as soon as possible
- Take steps to remediate any damage

Identify issues in the plane's defense and detection systems, then airline can refine its safety procedures and better protect passengers

https://boostprosystems.com/2021/04/01/what-is-private-5g/
https://www.networkworld.com/article/3254575/what-is-ipv6-and-why-aren-t-we-there-yet.html

Using AI To Refine and Evolve Towards Resilience

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Resilience Analysis and Metrics

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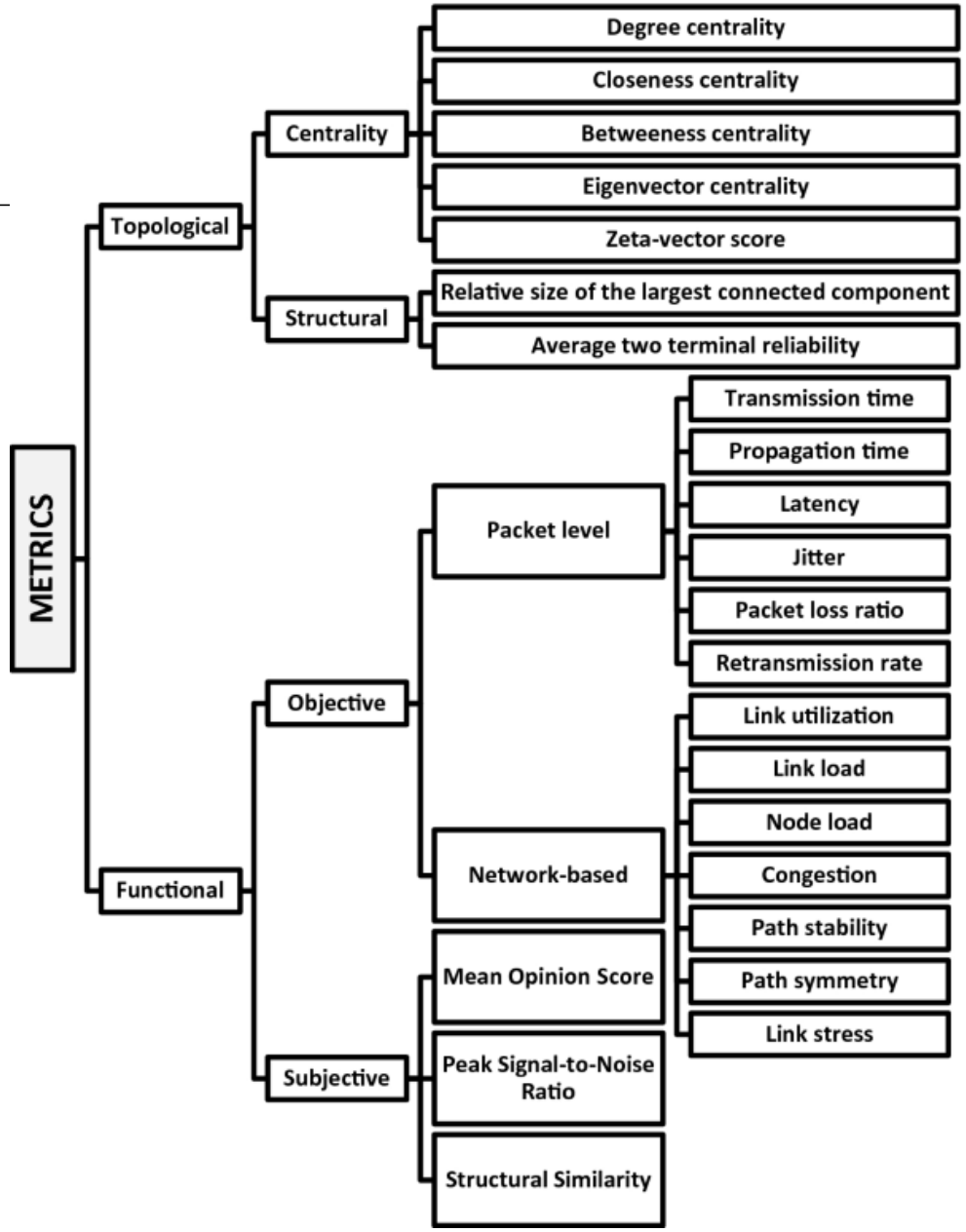
Metrics are required to

- Measure and quantify network resilience and understand it
- Evaluate the impact of events
- Improve the performance of the system on the long term

Challenges

- Most metrics are application- and domain-specific
- Evaluate the ability of the network to continue providing an acceptable level of service
  - What exactly is this service?
  - What is acceptable?

Some Example Metrics



Resilience Loss During a Challenge Event

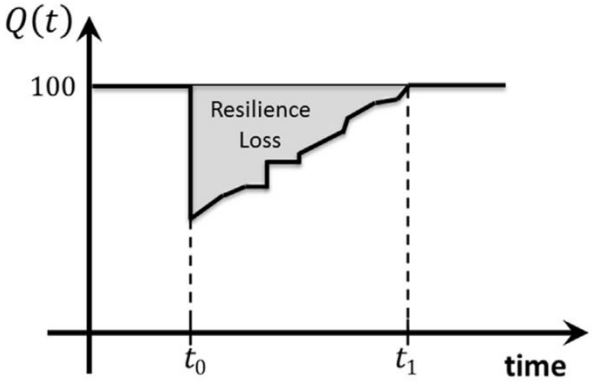
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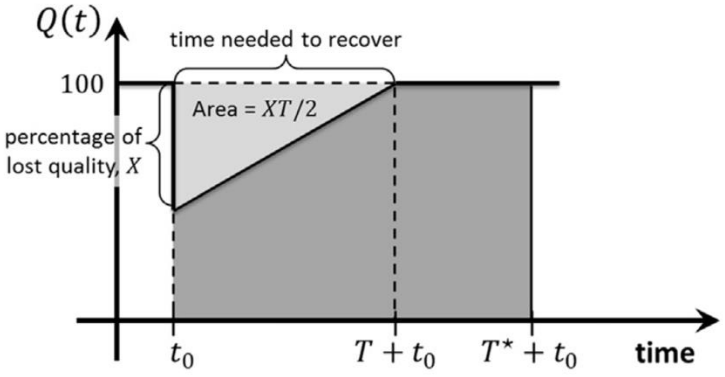
Resilience as the difference between

- The expected performance (== 100%)
- The actual performance

Accurate Concept - BUT hard to measure

- How to measure the performance degradation of a system?





Resilience as the difference between

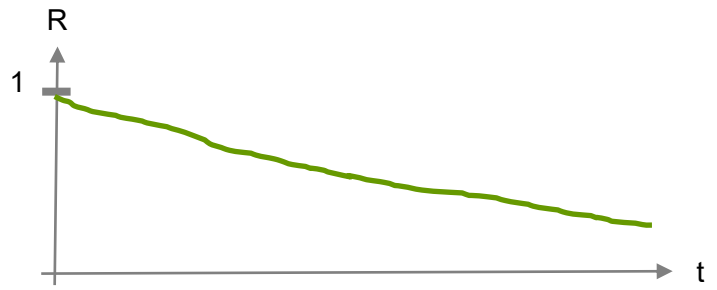
- The expected performance (== 100%)
- The performance at the start of the event
- As a linear interpolation

Easier to measure, but still...

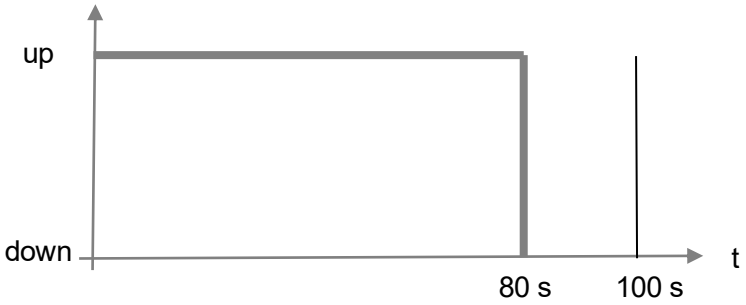
Reliability

Definition

- Captures if a system or service remains operable for a specified period of time (e.g., $[0, t]$)
- Measures the continuity of a service



Formula: $R = \frac{MTTF}{T}$



Based on the Mean Time To Failure (MTTF) and the time horizon T

- Usually, T is application dependent
- If an application is non-interruptable, continuity is essential

time interval: 100 s
reliability: 0.8 (80%)

MTTF: 80 sec
MTTR: 20 sec

Availability

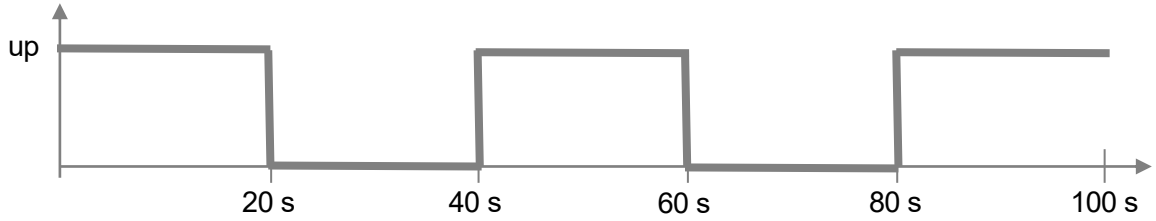
Definition

- Probability that a system or service will be operable when needed
- Measures the readiness for usage of a service

Formula: $A = \frac{MTTF}{MTTF+MTTR}$

Based on the Mean Time To Failure (MTTF) and the Mean Time To Recovery (MTTR)

- Is application independent, only based on failure and recovery times
- A similar concept is often utilized in Service-Level Agreements (SLAs)



time interval: 100 s
availability: 0.5
reliability: 0.2 (20%)

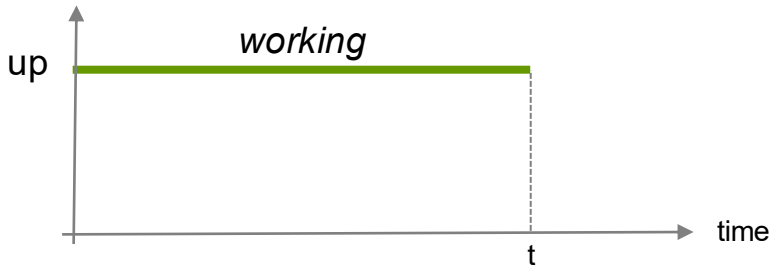
MTTF: 20 sec
MTTR: 20 sec

Reliability vs. Availability

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Reliability

expresses the probability that a system or service will be working **until** the time t



Availability

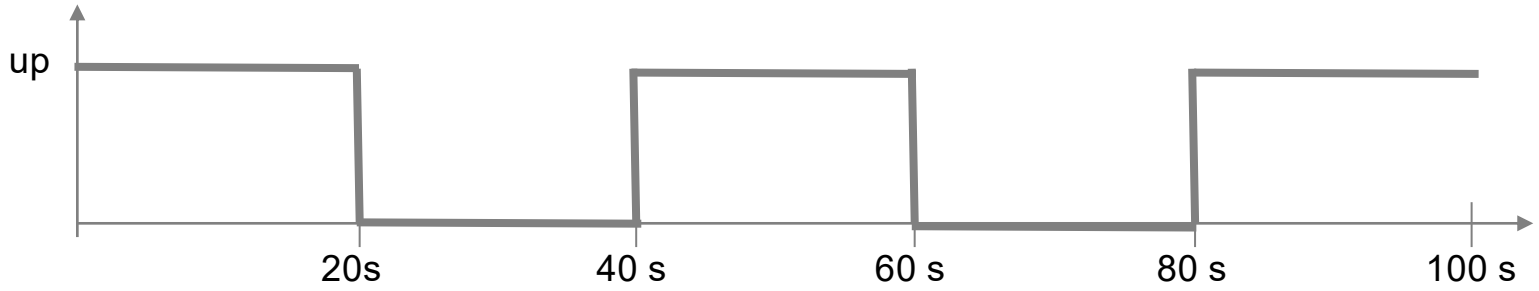
expresses the probability that a system or service will be working **at** the time t



Reliability vs. Availability

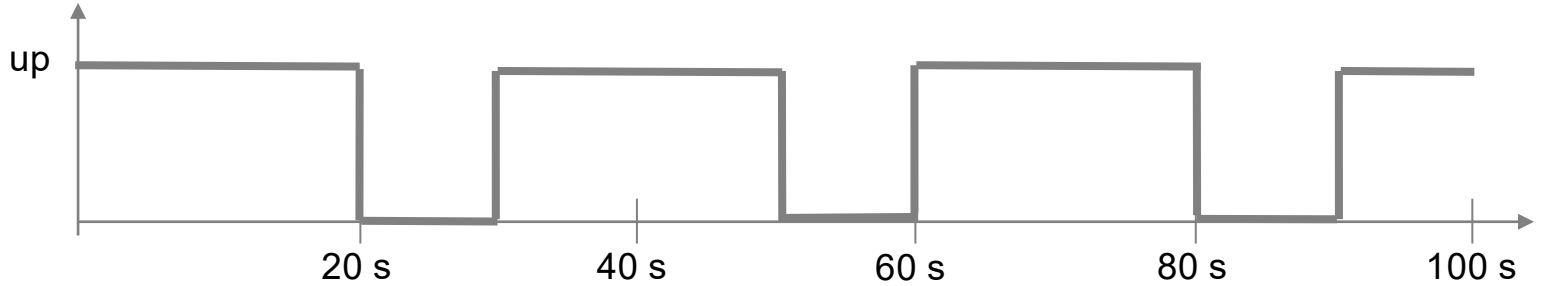
Reliability and availability are tightly related, but they are not the same!

Example 1:



time interval:100 s, **reliability**: 0.2, MTTF: 20 s, MTTR: 20 s, $A_{[0,100]}$: 0.5

Example 2:

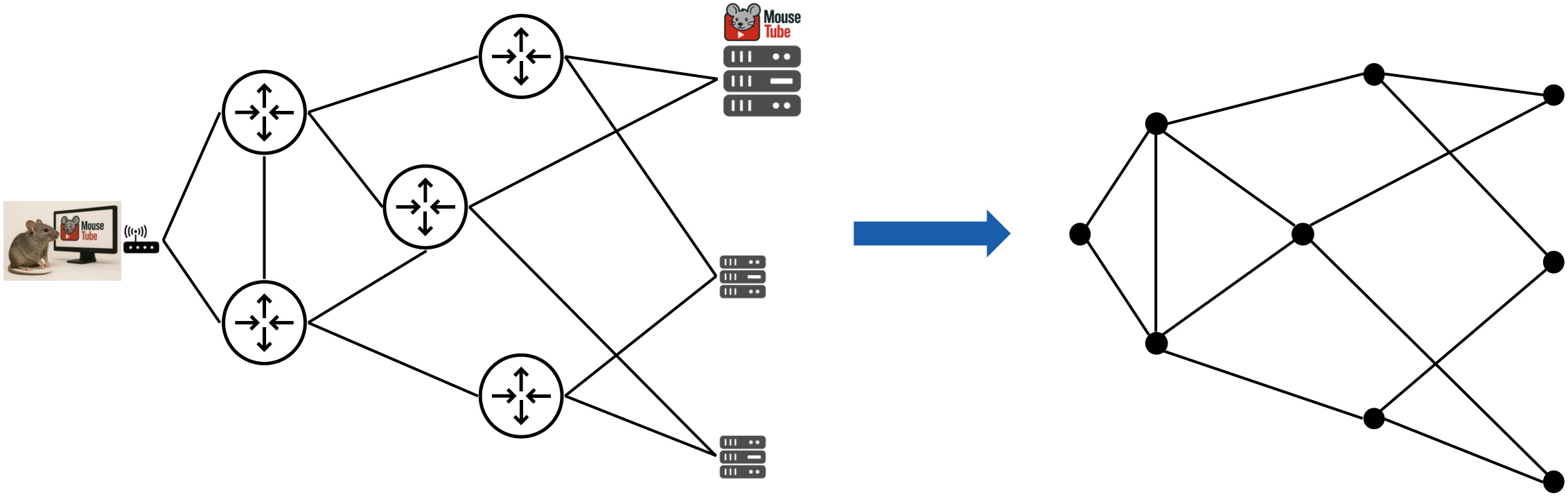


time interval:100 s, **reliability**: 0.2, MTTF: 20 s, MTTR: 10 s, $A_{[0,100]}$: 0.67

Graph Metrics

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Communication networks are graphs …



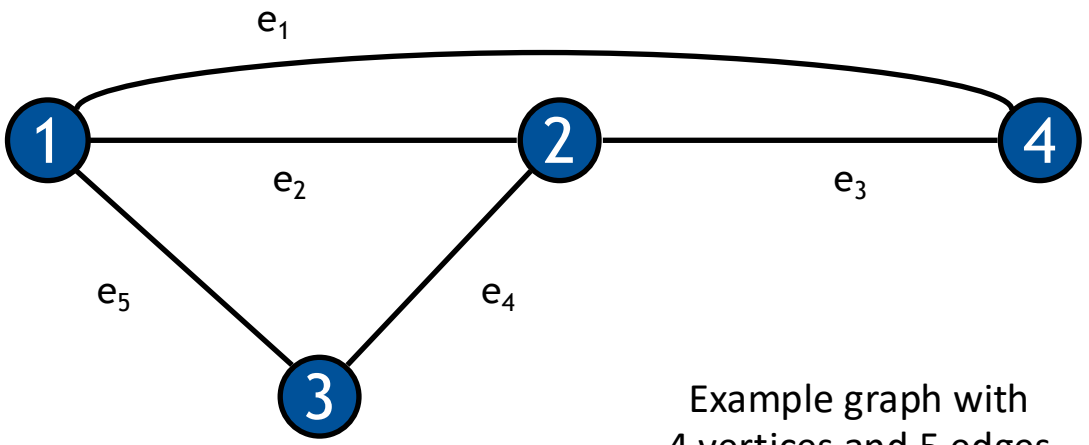
… so, graph metrics can be used to assess their resilience!

What Is a Graph?

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Definition:

**Graph *G = (V, E)* consists of two finite sets,
set *V* of vertices (nodes)
and set *E* of edges (arcs).**



Example graph with
4 vertices and 5 edges

Side note:

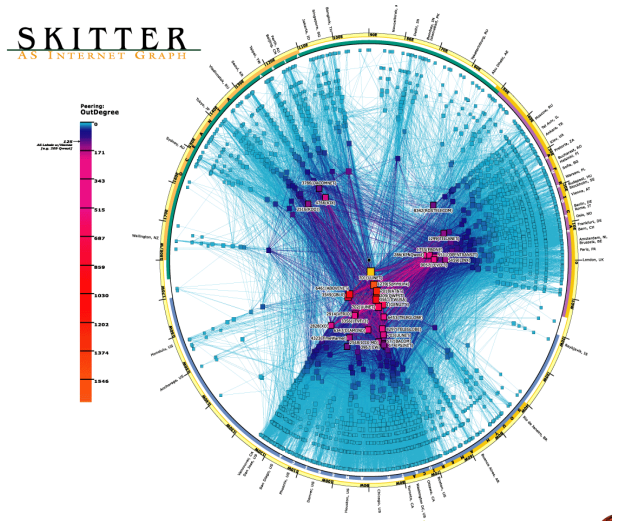
Edges can have (multiple) “weights” $w : E \rightarrow R$

The Internet

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Faloutsos et al. study from 99: Internet topology examined in 1998

- AS-level topology
- During 1998 Internet grew by 45%



Motivation:

- What does the Internet look like?
- Are there any topological properties that don't change over time?
- How to generate Internet-like graphs for simulations?

Four key properties found; each follows a power-law;
Sort nodes according to their (out)degree

- Out degree of a node is proportional to its rank to the power of a constant
- Number of nodes with same out degree is proportional to the out degree to the power of a constant
- Eigenvalues of a graph are proportional to the order to the power of a constant
- Total number of pairs of nodes within a distance d is proportional to d to the power of a constant

Worst-Case Connectivity Metrics

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Balanced (vertex) cut

- Minimum number of edges (vertices) removed to achieve a network fragmentation into two equal components

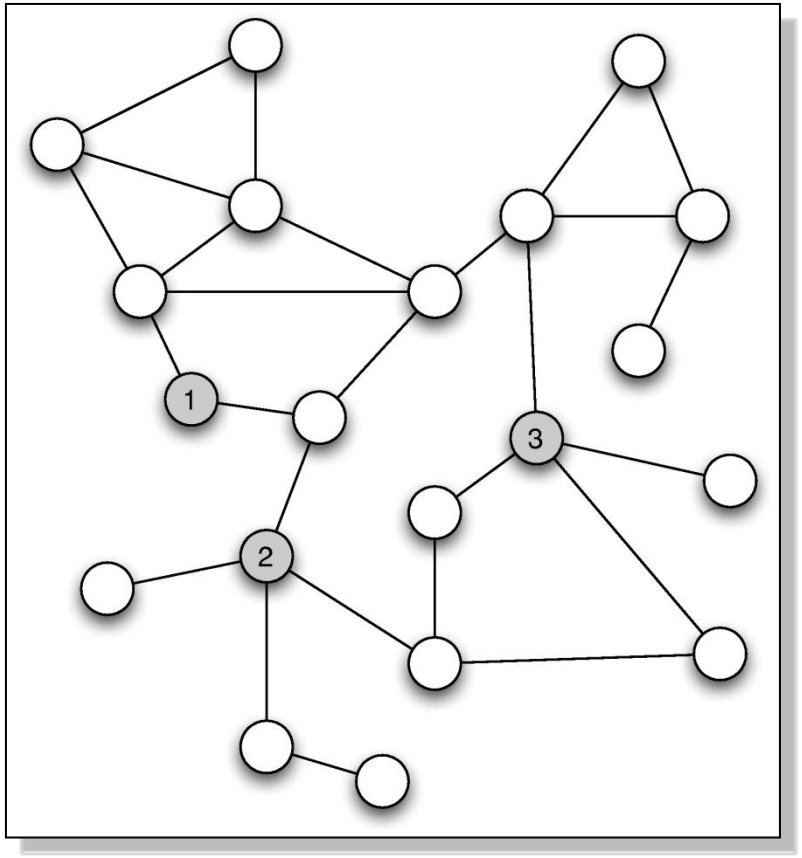
Maximum Isolated Component Size

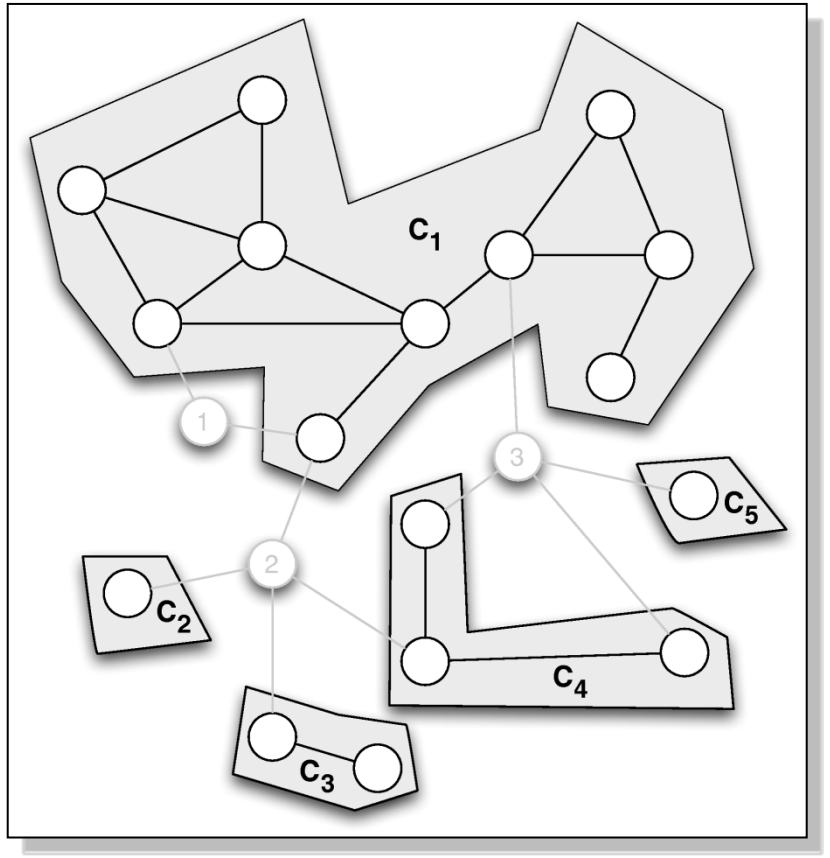
Average Isolated Component Size

„Point of Rupture“

- Minimum number of nodes the removal of which causes the network to fragment into components, with |giant component| < 0.5*|V|

Worst-Case Connectivity Metrics: Example





Resilience of Scale-Free Networks

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Experiment: take network of 10000 nodes (random and power-law) and remove nodes randomly

Random graph:

- Take out 5% of nodes: Maximum isolated component: 9000 nodes
- Take out 18% of nodes: Maximum isolated component: 100, all components between 1 and 100 nodes
- Take out 45% of nodes: Only groups of 1 or 2 survive

Power-law graph:

- Take out 5% of nodes: Only isolated nodes break off
- Take out 18% of nodes: Maximum isolated: 8000 nodes
- Take out 45% of nodes: Large cluster persists, fragments small

Networks with power law exponent < 3 are very robust against random node failures

- ONLY true for random failures!

Properties of Scale-Free Networks

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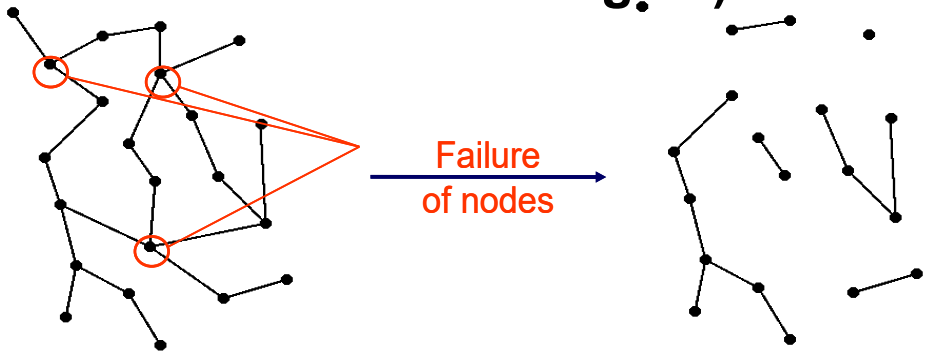
Fault tolerance / robustness against random failures = important property of networks with scale-free degree distribution

- Remove randomly chosen vertex v from: In SF it is a low-degree vertex with high probability => damage to the network small

But very sensitive to attacks

- Adversary removes highest degree vertices first => the network quickly decomposes into very small components

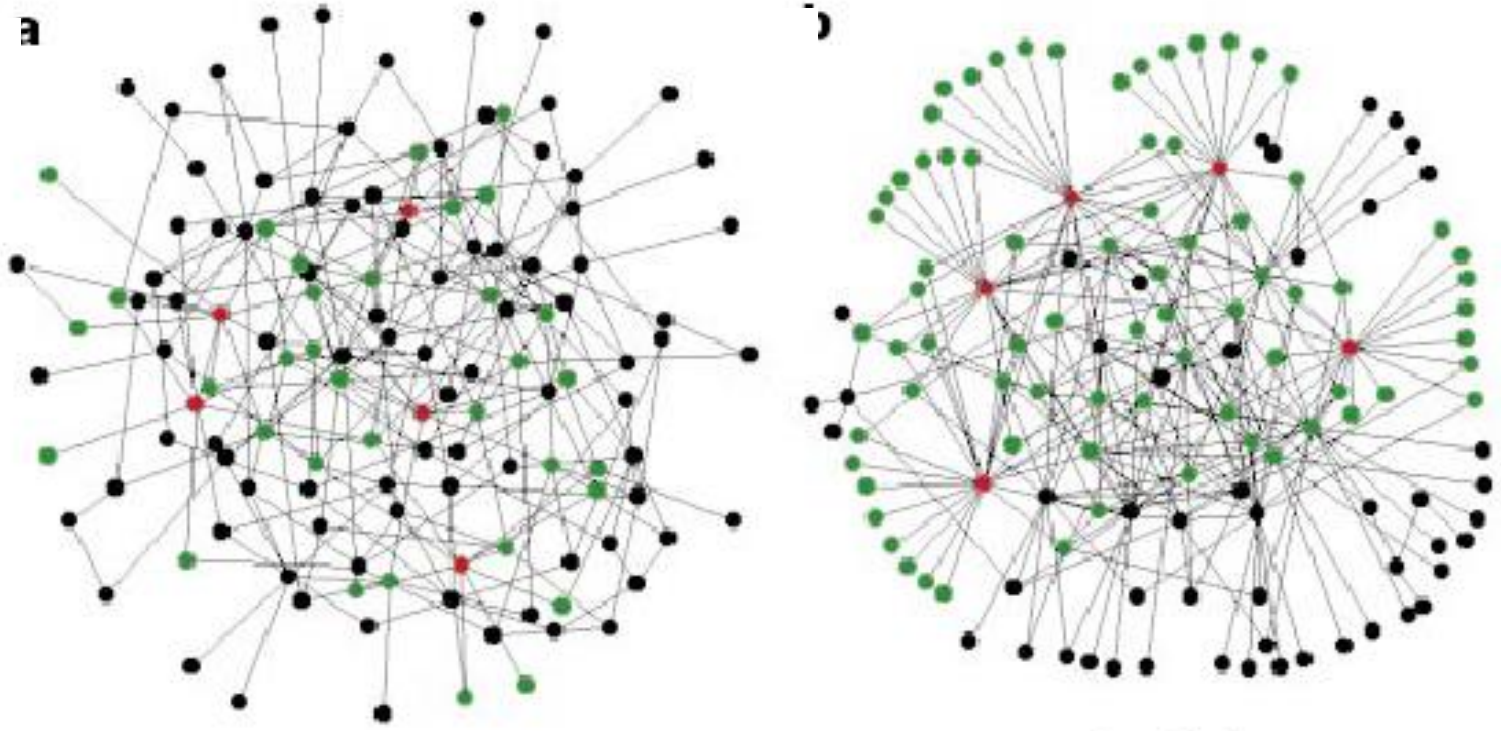
Note: random graphs are not robust to random failures, but not sensitive to attacks either (because all vertices more or less have the same degree)



Resilience of Scale-Free Networks

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Random failures vs. directed attacks



Random Graph

„Power Law“ Graph

Chaos Engineering

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"Chaos Engineering is the discipline of experimenting on a system in order to build confidence in the system's capability to withstand turbulent conditions in production."



Chaos Engineering

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Beginnings

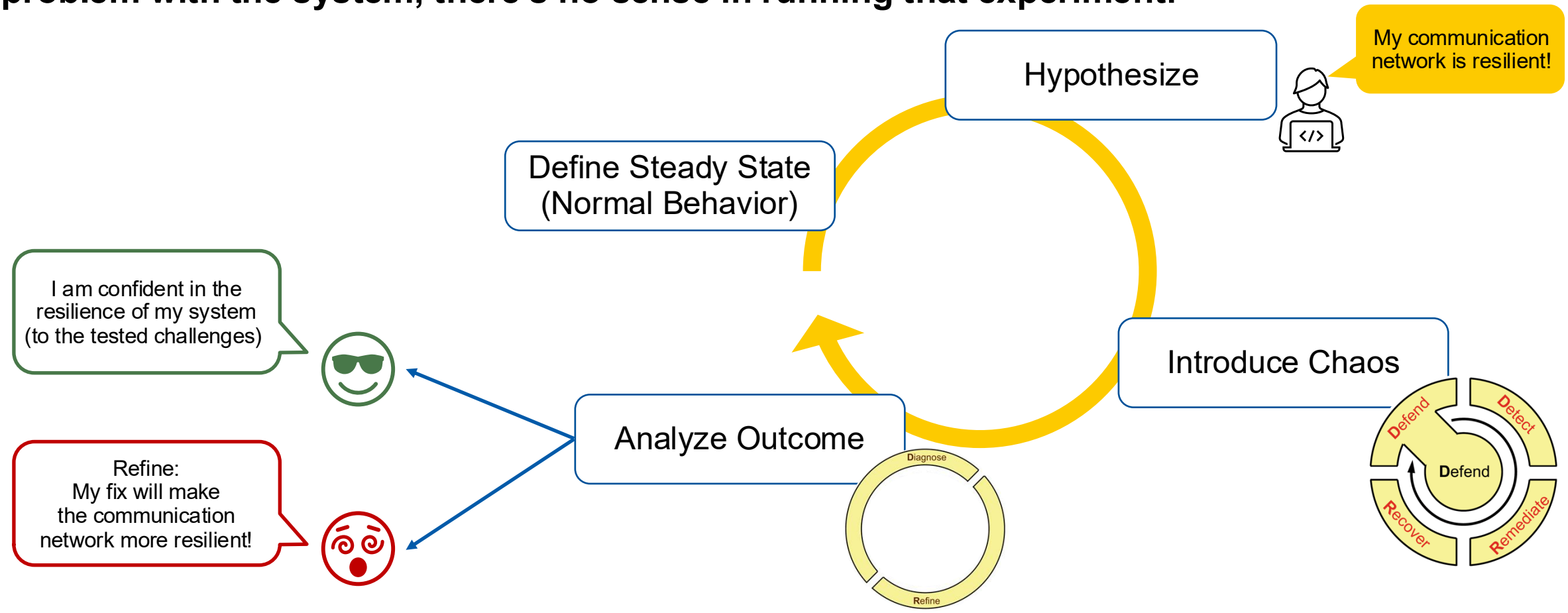
- Term “Chaos Engineering” coined by Netflix
  - Moved into the cloud in 2008
  - Invented *Chaos Monkey* tool to turn off services in the production environment

Resiliency Testing (in Production!)

- Trigger failure scenarios and challenge the system on purpose
- Observe system behavior and uncover unknown weaknesses
- Learn from outages and refine system to increase resilience

Chaos Engineering – Chaos Experiments

If you are certain that a Chaos Engineering experiment will lead to a significant problem with the system, there's no sense in running that experiment!



Chaos Engineering – Principles of Chaos

